

AVISHKAR MAHESH PAWAR

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Professional Summary

Computer Science student with strong foundation in systems programming and distributed database internals. Built a complete distributed database (MiniDB) from scratch in C++ spanning all 6 phases: B+ Tree indexing, MVCC concurrency control, Raft consensus, full query processing with JOINS, sharding, and HTTP monitoring. Proficient in low-level optimization, memory management, and concurrent programming. GATE 2026 qualified (AIR 3124) with 9.24 CGPA.

Education

MIT Academy of Engineering, Pune

2022 - 2026

B.Tech. - Computer Engineering — CGPA: 9.24/10

Projects

MiniDB — Distributed Database with ACID Transactions

C++17, Boost.Asio, Protocol Buffers, CMake

- Engineered a fully-featured distributed key-value database from scratch in C++17 (all 6 phases complete) supporting ACID transactions with MVCC concurrency control, handling 10,000+ ops/sec with <1ms latency
- Implemented B+ Tree indexing with LRU-based buffer pool management, reducing disk I/O by 60% through intelligent page caching and prefetching
- Built complete query processing engine with SQL-like parser supporting JOINS, and designed custom binary protocol over TCP using Boost.Asio with connection pooling

Event Streaming Platform

Java 17, Spring Boot, Netty, Apache Kafka Protocol

- Designed partition replication protocol with leader-follower architecture for fault tolerance and high availability
- Implemented offset management with configurable retention policies and compaction strategies
- Built cluster coordination using ZooKeeper for leader election and metadata management

AI-Powered Vector Database

Python 3.11, FastAPI, NumPy, Numba

- Implemented graph-based indexing (HNSW) with hierarchical layers for sub-linear search complexity $O(\log n)$ on 1,000,000+ vectors
- Designed sharding mechanism with consistent hashing and write-ahead logging for horizontal scalability and crash recovery
- Built RESTful API using FastAPI with async endpoints handling 1000+ concurrent requests

Technical Skills

Languages: C++, Java, Python, JavaScript, TypeScript

Frameworks: Spring Boot, FastAPI, React, Node.js, Express.js

AI/ML: TensorFlow, PyTorch, OpenAI API, LangChain, NumPy, Numba

Databases: PostgreSQL, MySQL, MongoDB, Redis, Custom Database Implementations

Tools: Git, Docker, Linux, AWS, CMake, Maven, Prometheus

Distributed Systems: Kafka, gRPC, Protocol Buffers, Raft Consensus, Consistent Hashing

CS Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Distributed Systems, System Design

Experience

OSMOS

Jul 2025 - Present

Software Developer

- Worked on backend services using Node.js and developed applications using Python
- Contributed to product development from scratch, including designing, implementing, and testing core features

Core-Decimal Solutions

Feb 2025 - Aug 2025

Software Developer

- Contributed to full-stack project using Vue.js, Express.js, and Sequelize with MVC architecture
- Developed reusable Vue.js components and integrated RESTful APIs with Axios

Achievements & Certifications

GATE 2026: Secured AIR 3124 with GATE Score 601 and 51.45/100 marks in CS — 3x rank improvement from GATE 2025 (AIR 9860)

GATE 2025: Qualified with AIR 9860, GATE Score: 453, Marks: 39.32/100

First Place in Technodium 25: Won first place in technical competition

Certifications: CCNA: Introduction to Networks, CCNA: Switching, Routing, and Wireless Essentials, CCNA: Enterprise Networking, Security, and Automation

Extra-Curricular Activities

Outdoor Sports & Games: Active participant in outdoor sports and recreational activities